Combinatorial Algorithms (Math 482), 2026 Spring   Problem Set 1

Due by Wednesday, January 28, 2026.

1. **Big-Oh notation.** Find all pairs of functions below such that $f(n) \in \Theta(g(n))$.

$$n^2 + n\log n + 3, \quad 32(n+1)(n-1), \quad n^2 \log^2 n, \quad \frac{5n^2}{\log n}(\log n + 2)^3, \quad \sqrt{n}(n^{3/2}+2).$$

2. **Ranking asymptotic growth rates.** Sort the following 8 functions in ascending order by asymptotic growth rate. That is, any two consecutive functions, $f(n)$ and $g(n)$, should satisfy $f(n) \in O(g(n))$.

$$n^4, \quad 2^{\sqrt{\log n}}, \quad n\log^2 n, \quad 2^n, \quad (n+1)!, \quad 2^n n!, \quad n2^n, \quad n\log n.$$

3. **Analysis of snippets.** The following four algorithms run on an $n \times n$ array $A[1..n][1..n]$. Determine the worst-case asymptotic running time of each in terms of n.

(a)

```
begin
    M ⟵ 0
    i ⟵ 1
    while i ≤ n and M < A[i][i] do
        for j ← i to n do
            if M < A[i][j] then
                M ⟵ A[i][j]
        i ⟵ i + j
```

(b)

```
begin
    M ⟵ 0
    i ⟵ 1
    while i ≤ n do
        j ⟵ 2
        while j ≤ n do
            if M < A[i][j] then
                M ⟵ A[i][j]
            j ⟵ j + i
        i ⟵ i + 1
```

(c)

```
begin
    M ⟵ 0
    i ⟵ 1
    while i ≤ n do
        j ⟵ 2
        while j ≤ n do
            if M < A[i][j] then
                M ⟵ A[i][j]
            j ⟵ j^2
        i ⟵ 2 * i
```

(d)

```
begin
    M ⟵ 0
    i ⟵ 1
    while i ≤ n do
        j ⟵ 2
        while j ≤ n do
            if M < A[i][j] then
                M ⟵ A[i][j]
            j ⟵ i + j
        i ⟵ 2 * i
```

4. Recall the interval scheduling problem.

   We are given n intervals $(A[1], B[1]), \ldots, (A[n], B[n])$. Find a maximum conflict-free set of intervals. Two intervals $(A[i], B[i])$ and $(A[j], B[j])$ are not in conflict if they do not overlap, that is, $B[i] \leq A[j]$ or $B[j] \leq A[i]$.

   (a) Find a solution for the instance $(1, 3)$, $(2, 5)$, $(4, 7)$, $(6, 9)$, $(7, 8)$. Prove that your answer is correct.

   (b) Maximum vs maximal. A conflict-free set of intervals is *maximum* if it consists of the largest possible number of intervals; and it is *maximal* if any additional intervals would create a conflict. Construct an input for the problem, where a *maximal* conflict-free sets of intervals is not a *maximum* conflict-free set.

   (c) Construct an input of $n = 6$ intervals where the solution consists of one interval. How many different optimal solutions are there in the instance that you have constructed?

   (d) Construct an input of $n = 6$ intervals where the solution consists of two intervals. How many different optimal solutions are there in the instance that you have constructed?